Regularization and Validation

6

Pre

CS 3244
Machine Learning

NUS National University of Singapore | Computing

Recap and Forecast

Bias: Cost due to using $\mathcal{H}$ to approximate f over any dataset.

$$\mathbb{E}_x\left(\bar{h}(x) - f(x)\right)^2$$

Variance: Cost due to the specific dataset $\mathcal{D}$.

$$\mathbb{E}_{\mathcal{D},x}\left[\left(h_{\mathcal{D}}(x) - \bar{h}(x)\right)^2\right]$$

Stochastic Noise: Uncontrollable noise.

$$\mathbb{E}_{\epsilon,x}\left(\epsilon(x)\right)^2$$

Match the 'model complexity' to the data resources, not to the target complexity.

(Soft) SVM: Optimal linear classifier determined by support vectors (with tolerance for errors)

Learning Outcomes for this week

- Understand **Regularization** as a means of restraining the model.
- Choose appropriate doses of regularization for a model.
- Understand and execute **Validation**, as a reality check by peeking (at the bottom line).
- Understand the different forms extending validation to encompass additional estimation.
- Understand how validation and regularization complement each other and their roles in affecting learning.

Regularization

CS3244 Machine Learning

NUS Department of Computer Science
School of Computing

Two Cures

In one form or another, $J_{test}(h) = J_{train}(h) + \text{overfit penalty}$

1. **Regularization**: Restrain the model

$$J_{test}(h) = J_{train}(h) + \text{overfit penalty}$$

Regularization estimates this quantity

2. **Validation**: Reality check by peeking (at the bottom line)

$$J_{test}(h) = J_{train}(h) + \text{overfit penalty}$$

Restraining the model

Regularization

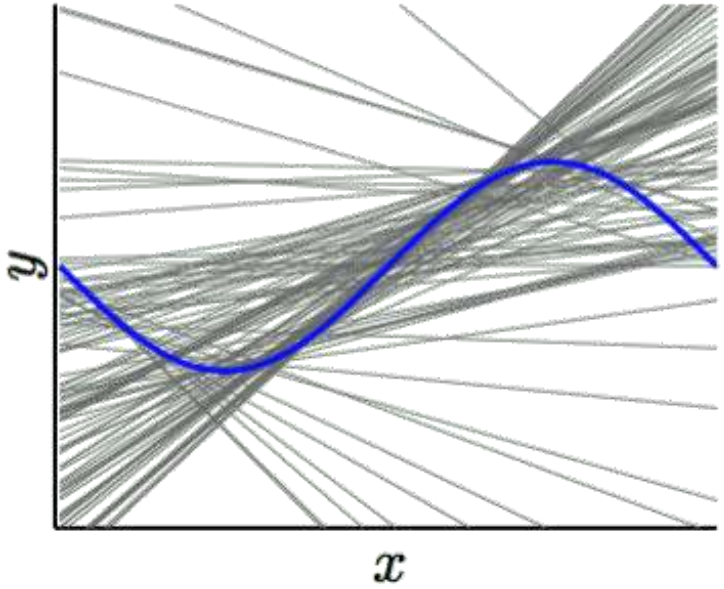
- What is it: A cure for our tendency to fit (get distracted by) the noise, hence improving J_{test}.

- How does it work?
  By constraining the model so that we cannot fit the noise

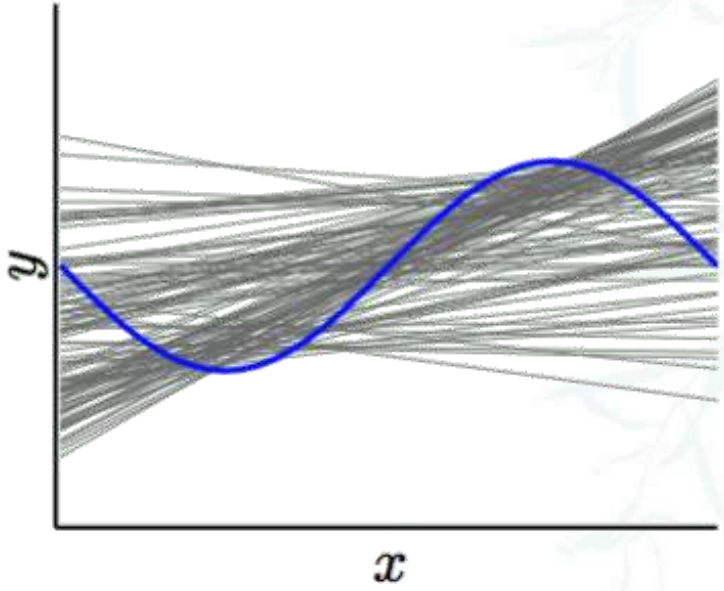
- Side effect: if we cannot fit the noise, maybe we cannot fit the signal f.

A familiar example

Without regularization



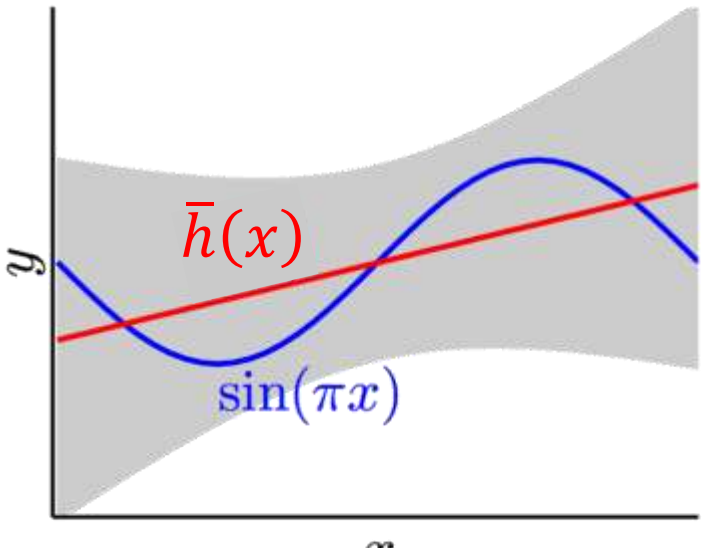
With regularization



Constrain weights to be a bit smaller

Bias goes up a little

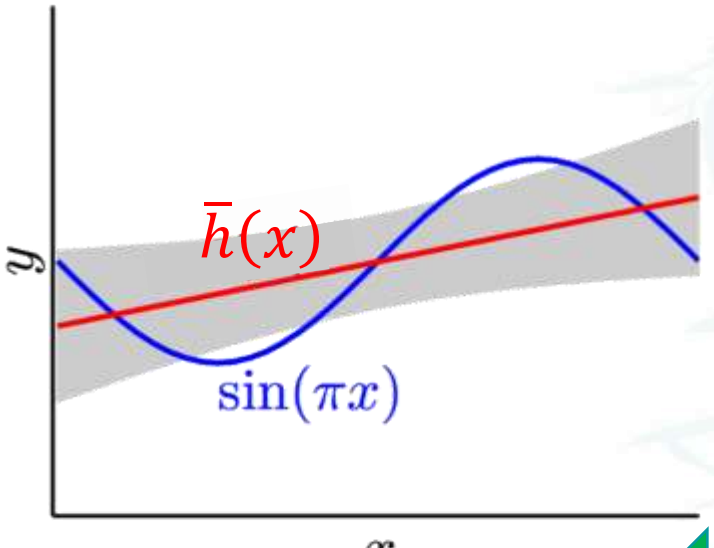
Without regularization



Bias = 0.21

Variance = 1.69

With regularization



Bias = 0.23

Variance = 0.33

Side Effect

Constraining the model: $\mathcal{H}_2$ versus $\mathcal{H}_{10}$

$$\mathcal{H}_{10} = \{h(x) = \theta_0 + \theta_1 x + \cdots + \theta_{10} x^{10}\}$$

$$\mathcal{H}_2 = \{h(x) = \theta_0 + \theta_1 x + \cdots + \theta_{10} x^{10}\}$$

such that $\theta_3 = \theta_4 = \cdots = \theta_{10} = 0$

A "hard" order constraint that sets some weights to zero.

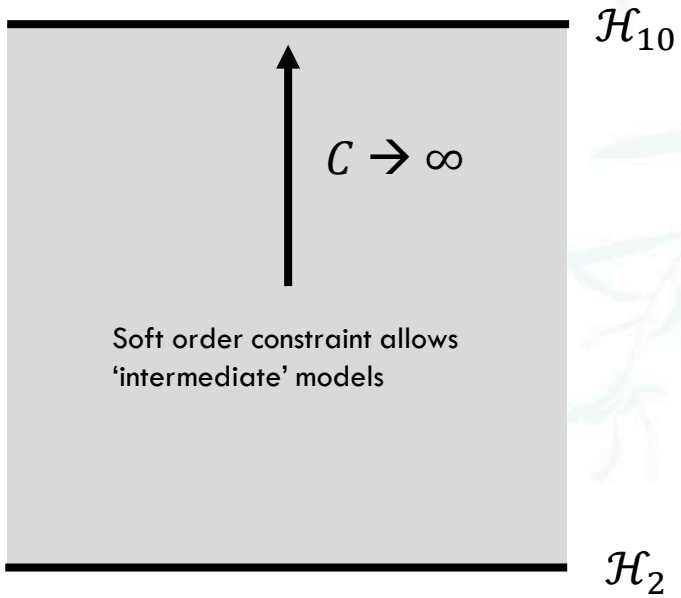
$$\mathcal{H}_2 \subset \mathcal{H}_{10}$$

Soft Order Constraint

Don't set weights explicitly to zero.

Reuse optimization by giving a budget and let the learning choose.

$$\sum_{q=0}^{Q} \theta_q^2 \leq C$$



Soft Order Constrained Model

$$\mathcal{H}_{10} = \{h(x) = \theta_0 + \theta_1 x + \cdots + \theta_{10} x^{10}\}$$

$$\mathcal{H}_C = \{h(x) = \theta_0 + \theta_1 x + \cdots + \theta_{10} x^{10}\}$$

such that $\sum_{q=0}^{10} \theta_q^2 \leq C$

H_2 and C=2 are unrelated!

$$\mathcal{H}_2 = \{h(x) = \theta_0 + \theta_1 x + \cdots + \theta_{10} x^{10}\}$$

such that $\theta_3 = \theta_4 = \cdots = \theta_{10} = 0$

$\mathcal{H}$ is larger

$\mathcal{H}$ is smaller

Linear Regression Comparison

<u>Unconstrained</u>

$$\min J_{train} = \frac{1}{m}\sum_{j=1}^{m}\left(\theta^\top x^{(j)} - y^{(j)}\right)^2$$

$$\min J_{train} = \frac{1}{m}(X\theta - y)(X\theta - y)$$

$$\theta_{lin} = (X^\top X)^{-1}X^\top y$$

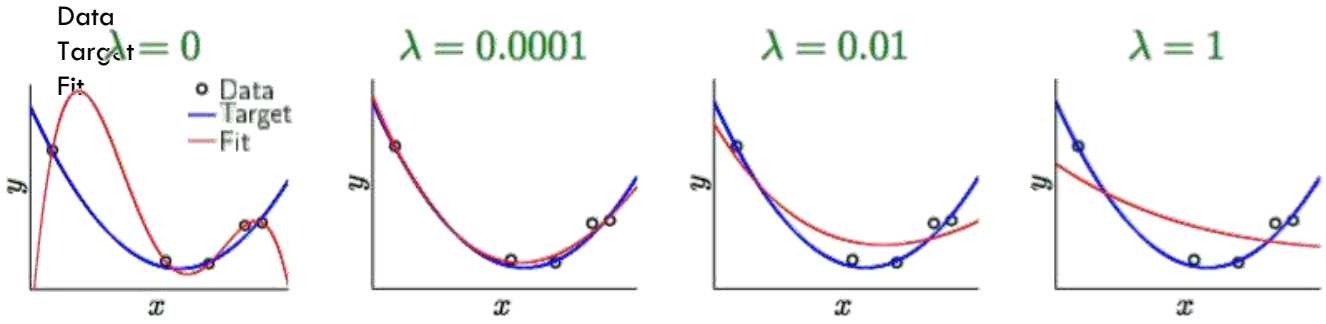
<u>Constrained</u>

$$\min J_{train} = \frac{1}{m}(X\theta - y)(X\theta - y),$$

subject to: $\theta^\top\theta \leq C$

Take the right λ amount of medicine

Minimizing $J_{train}(\theta) + \frac{\lambda}{m}\theta^{\top}\theta$ for different λ's:



overfitting $\rightarrow$ $C \rightarrow 0$, $\lambda \rightarrow \infty$ $\rightarrow$ underfitting

Quick Question: What happens θ to in the limit as $\lambda \rightarrow \infty$?

Validation

CS3244 Machine Learning

NUS National University of Singapore | Department of Computer Science
School of Computing

Two Cures

In one form or another, $J_{test}(h) = J_{train}(h) + \text{overfit penalty}$

1. **Regularization**: Restrain the model

$$J_{test}(h) = J_{train}(h) + \text{overfit penalty}$$

2. **Validation**: Reality check by peeking (at the bottom line)

$$J_{test}(h) = J_{train}(h) + \text{overfit penalty}$$

Validation estimates this quantity

Analyzing the estimate

On a test <u>point</u> (x, y), the cost $j(h_\theta(x), y)$ is:

Squared error: $(h_\theta(x) - y)^2$

Binary error: $[[h_\theta(x) \neq y]]$

$$\mathbb{E}[j(h_\theta(x), y)] = J_{test}(h_\theta)$$

$$Var[j(h_\theta(x), y)] = \sigma^2$$

From a point to a set

On a validation <u>set</u> $(x^{(1)}, y^{(1)}), \dots, (x^{(K)}, y^{(K)})$,
the cost is $J_{val}(h) = \frac{1}{K}\sum_{k=1}^{K} j(h(x^{(k)}), y^{(k)})$

$$\mathbb{E}[J_{val}(h)] = \frac{1}{K}\sum_{k=1}^{K} \mathbb{E}[j(h(x^{(k)}), y^{(k)})] = J_{test}(h)$$

$$Var[J_{val}(h)] = \frac{1}{K^2}\sum_{k}^{K} Var[j(h(x^{(k)}), y^{(k)})] = \frac{\sigma^2}{K}$$

$$J_{val}(h) = J_{test}(h) \pm O\left(\frac{1}{\sqrt{K}}\right)$$

Standard Deviation

K is taken out of m

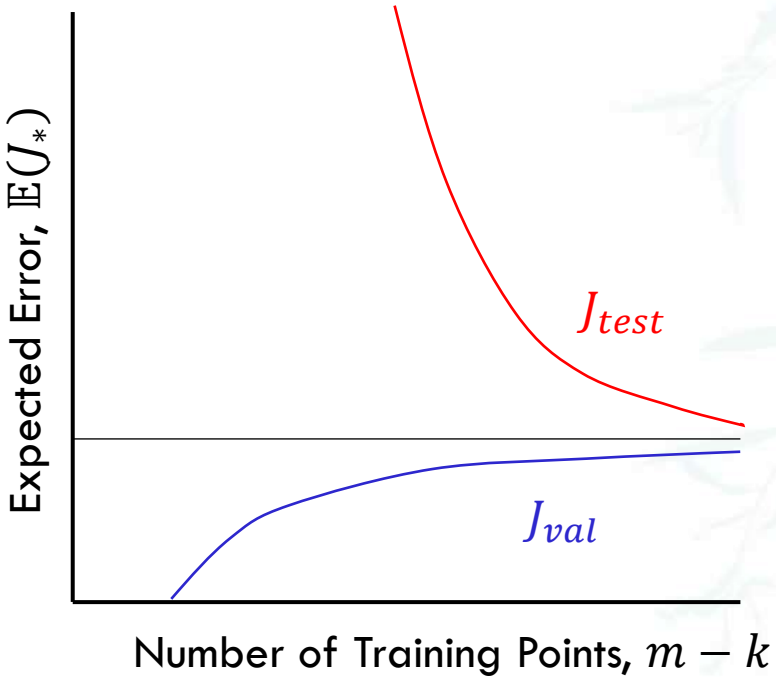
Given the data set $\mathcal{D}$, separate:

1. K points for validation
2. $m - K$ points for training

$O\left(\frac{1}{\sqrt{K}}\right)$:

Small K = bad estimate

Large K = ?



K is put back into m

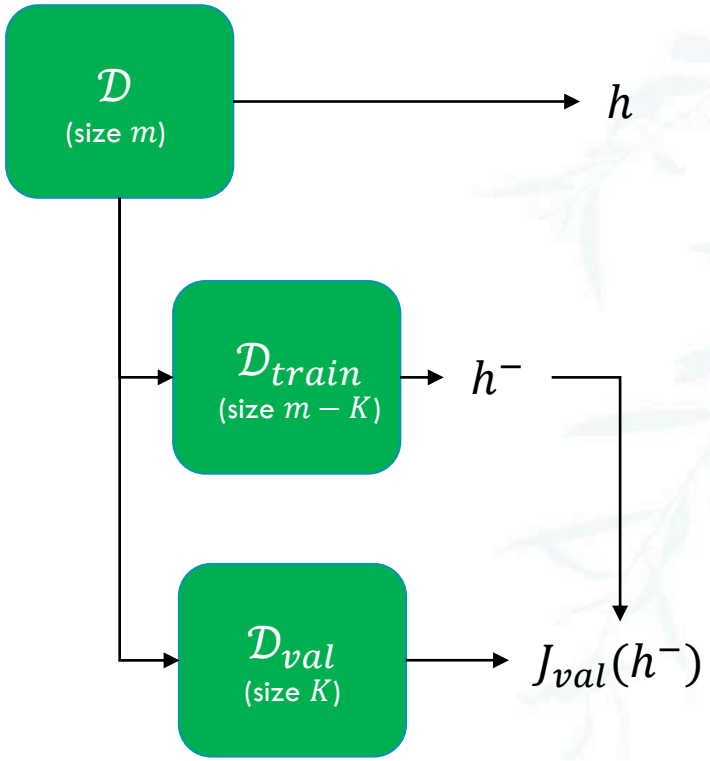
1. Train on $\mathcal{D}_{train}$ to yield h^-
2. Test h^- on $\mathcal{D}_{val}$ to yield J_{val}
3. Use cost J_{val} to estimate $J_{test}(h^-)$
4. Use h (not h^-!) in the end

Large K?

h^- trained on too few examples.

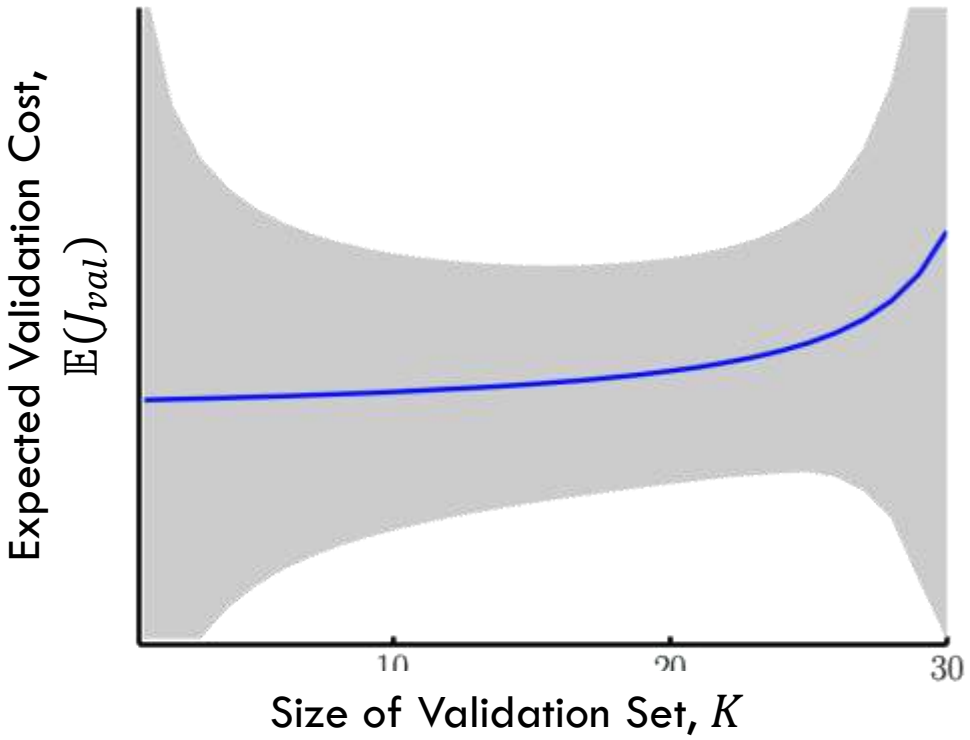
Leads to bad h^-, poor estimate.

Rule of Thumb: $K = \frac{m}{5}$



Expected Validation Error for $\mathcal{H}_2$

With $m = 40$, and noise level $= 0.4$



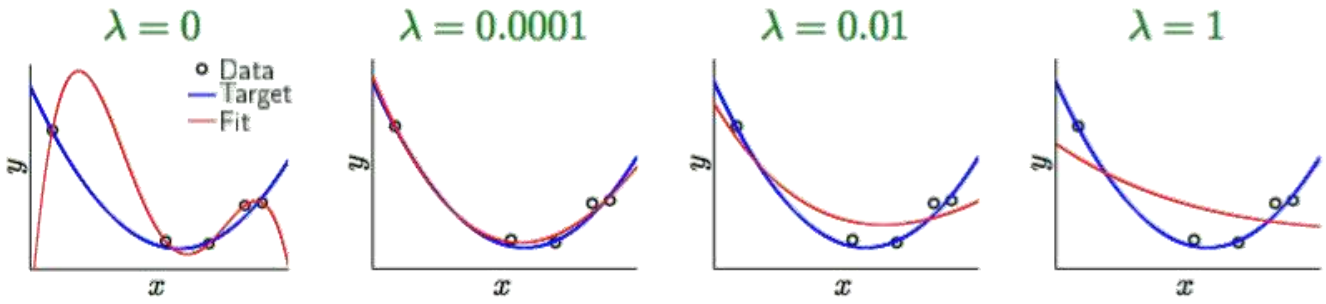
Cleanliness of Validation

CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

Why 'validation'?

Because $\mathcal{D}_{val}$ is used to make learning choices,
e.g. choosing the level of regularization to minimize J_{val}



If an estimate J_{test} affects learning:

The set is no longer a test set, it becomes a validation set!

What's the difference?

We know that the test set is unbiased.
But what about the validation set?

Q: Does the validation set have an optimistic or pessimistic bias?

Two hypotheses h_1 and h_2 with $J_{test}(h_1) = J_{test}(h_2) = 0.5$

Error estimates j_1 and j_2 uniform on $[0,1]$

We pick $h \in \{h_1, h_2\}$ by virtue of its $j = \min(j_1, j_2)$.

What then, is the value of $\mathbb{E}(j)$?

leading to bias.

What's the difference?

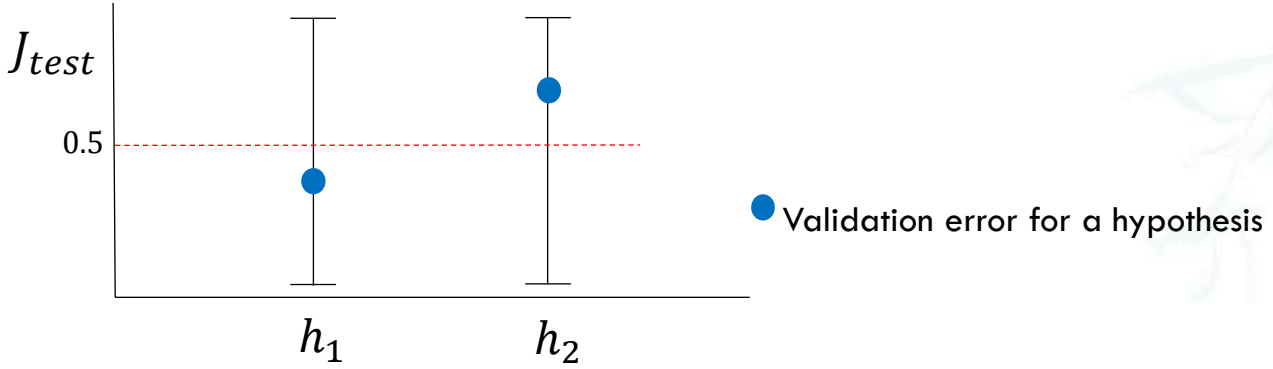
Two hypotheses h_1 and h_2 with $J_{test}(h_1) = J_{test}(h_2) = 0.5$

Error estimates j_1 and j_2 uniform on $[0,1]$

We pick $h \in \{h_1, h_2\}$ by virtue of its $j = \min(j_1, j_2)$.

What then, is the value of $\mathbb{E}(j)$?

$\mathbb{E}(j) < 0.5$, leading to an optimistic bias.



Hypothesis Selection: Using $\mathcal{D}_{val}$ more than once

Θ models $\mathcal{H}_1, \dots, \mathcal{H}_\Theta$

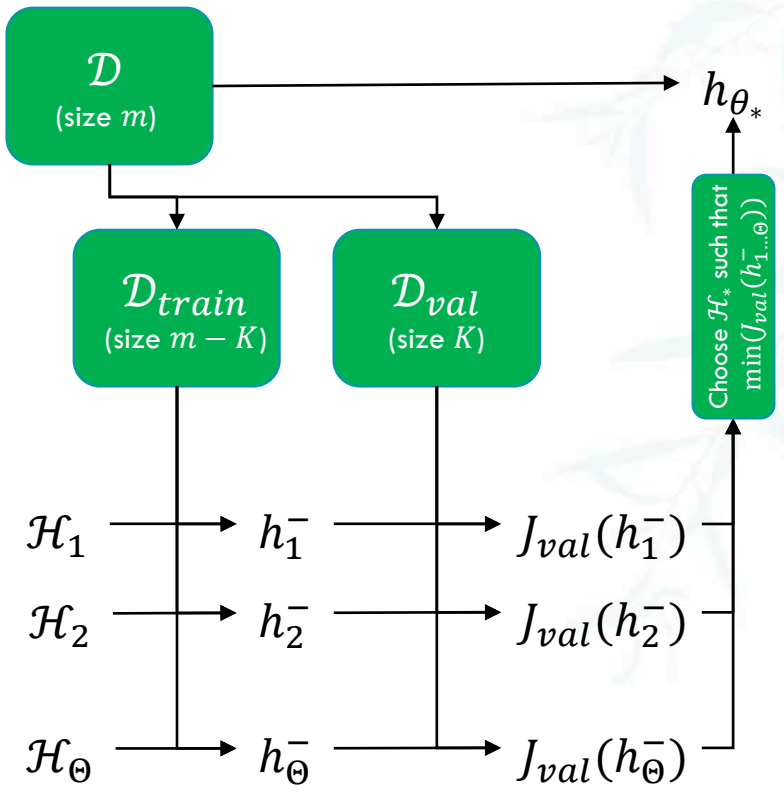
Use $\mathcal{D}_{train}$ to learn h_θ^- for each model

Evaluate h_θ^- using $\mathcal{D}_{val}$:

$$J_\theta = J_{val}(h_\theta^-); \quad \theta = 1, \dots, \Theta$$

Pick model $\theta = \theta_*$ with smallest J_θ

Uh oh!
Bias!



Data contamination

Error estimates: J_{train}, J_{test}, J_{val}

Contamination: optimistic (deceptive) bias in estimating J_{test}

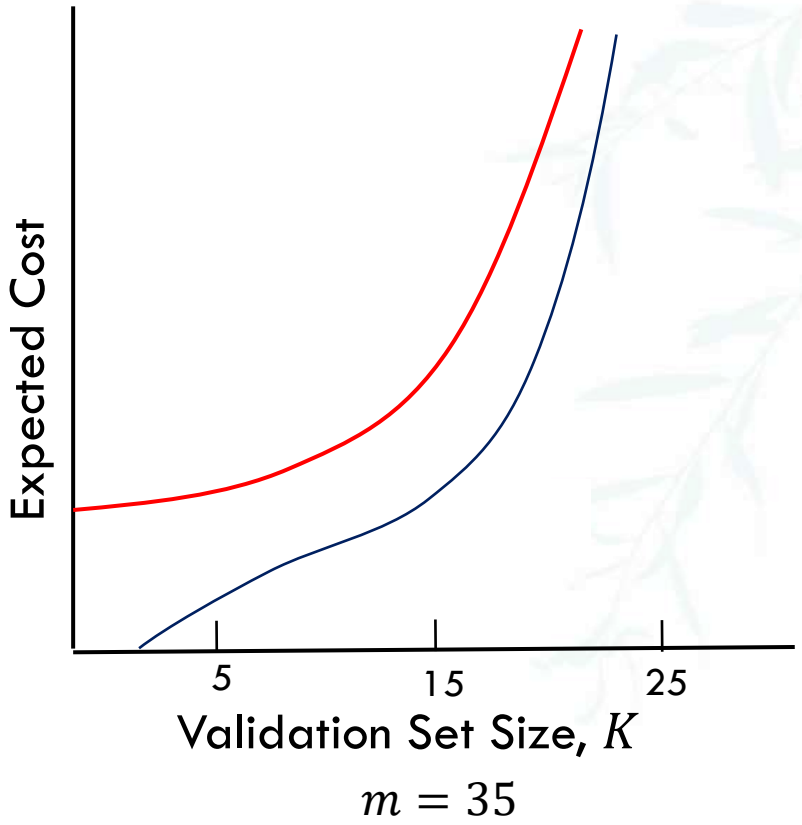
- Training set: totally contaminated
- Validation set: slightly contaminated
- Test set: totally 'clean'

Time Out: Check your understanding

We selected the model $\mathcal{H}_{\theta^*}$ using $\mathcal{D}_{val}$. We can think of validation as training among the set of Θ models.

$J_{val}(h_{\theta^*}^{-})$ is a biased estimate of $J_{test}(h_{\theta^*}^{-})$

<u>Quick Quiz:</u>

One curve is J_{val}, and the other J_{test}

1. Which curve is J_{val}?
2. Why are the curves going up?
3. Why do the curves get closer together?